Afriverse – Developer Infrastructure Accounts Setup

This document outlines the core infrastructure accounts to be created using the official **Afriverse email**. These tools will power our backend, hosting, authentication, and database for Version 1 (V1).

For now, we are starting lean with three essential platforms.

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1 MongoDB

https://www.mongodb.com/

What It Is

MongoDB is a cloud-based NoSQL database used to store and manage application data in flexible document format.

Why Afriverse Needs It

MongoDB will serve as our **primary database**. It will store:

- User accounts
- Authentication data
- Platform records
- Transactions (if applicable)
- Admin configurations (if applicable)
- Logs and system data

Since Afriverse will evolve and expand, MongoDB's flexible schema allows us to iterate quickly without rigid database constraints.

Why It's the Right Choice

- Works seamlessly with Node.js
- Scalable for future growth
- Secure cloud hosting (MongoDB Atlas)
- Free tier available for early-stage testing

Plan for V1

We will use **MongoDB Atlas Free Tier (M0)** during development and testing.
Upgrade only when traffic and data volume increase.

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2 Render (Backend Hosting Server)

https://render.com/

What It Is

Render is a cloud hosting platform used to deploy backend servers (Node.js, Express APIs, etc.).

Why Afriverse Needs It

Render will host:

- Our backend API
- Business logic
- Authentication system
- Database connections
- Webhooks and integrations

This ensures Afriverse is accessible online 24/7.

Pricing Structure

- **Free Plan – $0/month**
  - Good for V1 testing
  - Server sleeps after inactivity
  - Suitable for development phase
- **Starter Paid Plan – $7/month**
  - Always-on server
  - No cold starts
  - More stable for production
  - Recommended once V1 is validated

Plan for V1

We will start with the **Free Plan** to test and validate.
Once stable and ready for public users, we upgrade to the **$7/month plan**.

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3 Firebase

https://firebase.google.com/

What It Is

Firebase is a Backend-as-a-Service platform by Google providing authentication, analytics, notifications, and optional real-time capabilities.

Why Afriverse Needs It

We can use Firebase for:

- User authentication (Google login, email/password)
- Push notifications
- Analytics tracking
- File storage (images, documents)
- Real-time features (if required later)

Firebase accelerates development by reducing the need to build infrastructure from scratch.

Plan for V1

We will use the **Firebase Spark Plan (Free Tier)** during initial development.
Upgrade only if usage exceeds free limits.

4 Cloudinary (Media Storage & Optimization)

https://cloudinary.com/

What It Is

Cloudinary is a cloud-based media management platform used to store, optimize, and deliver images and videos efficiently.

Why Afriverse Needs It

Afriverse will likely require media handling such as:

- User profile images
- Platform banners
- Uploaded content

- Event or product visuals
- Marketing assets

Instead of storing images directly on our server (which increases load and cost), Cloudinary:

- Stores media securely in the cloud
- Automatically optimizes image size and quality
- Provides fast global delivery via CDN
- Reduces backend server strain

Why It's Important

- Improves website speed
- Enhances user experience
- Scales easily as media grows
- Reduces infrastructure complexity

Plan for V1

Use **Cloudinary Free Tier** for testing and early usage.
Upgrade only when storage or bandwidth increases significantly.

Afriverse V1 Infrastructure Strategy

Our approach:

- Build lean
- Keep costs minimal
- Validate quickly
- Upgrade only when necessary

Estimated Monthly Cost (V1 Testing Phase)

Tool	Plan	Cost
MongoDB	Free Tier	$0
Render	Free Tier	$0
Firebase	Free Tier	$0
Cloudinary	Free Tier	$0

Total (V1 Testing Phase): $0/month

After validation:

- Render > $7/month
- Others upgrade only if required

As Afriverse scales, we may later integrate:

- Cloudflare (Security & CDN)
- Stripe (Payments)
- Monitoring tools
- CI/CD automation
- Advanced analytics systems

These will be documented in the next infrastructure phase.